Lesson 9: Convex Optimization and Duality

Lagrangian Duality, KKT Conditions, and First-Order Methods

Session 9 of 102 | Phase I: Mathematical Foundations | 8h

Primary texts: Boyd & Vandenberghe Ch. 4–5, 9–11; Nesterov Ch. 1–4

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1 Introduction

Optimization is the computational engine beneath virtually every algorithm in machine learning and reinforcement learning. Policy gradient methods solve surrogate optimization problems at each iteration; value function approximation minimizes projected Bellman error; and constrained MDPs reduce to saddle-point problems whose structure is governed by duality theory.

This document develops the theory of convex optimization from first principles: the hierarchy of convex programs (Section 2), Lagrangian duality and the passage from primal to dual (Section 3), the Karush–Kuhn–Tucker conditions (Section 4), gradient descent and its convergence guarantees (Section 5), Nesterov's accelerated method (Section 6), proximal methods for non-smooth objectives (Section 7), and interior-point methods (Section 8). We close with exercises (Section 9) and a summary table of convergence rates (Section 10).

Throughout, we assume familiarity with real analysis (continuity, compactness) and linear algebra (eigenvalues, positive definiteness) at the level of the Phase 00 prerequisites.

2 Convex Optimization Problems

2.1 Standard Form

Definition 2.1 (Convex optimization problem)**.** A *convex optimization problem* in standard form is

$$\begin{aligned} \min_{x\in\mathbb{R}^n} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \le 0, \quad i = 1,\dots,m, \\ & a_j^\top x = b_j, \quad j = 1,\dots,p, \end{aligned} \tag{1}$$

where $f_0, f_1, \dots, f_m : \mathbb{R}^n \to \mathbb{R}$ are convex functions and the equality constraints are affine.

Definition 2.2 (Feasible set, optimal value)**.** The *feasible set* is $\mathcal{F} = \{x \in \mathbb{R}^n : f_i(x) \le 0,\ a_j^\top x = b_j\}$. The *optimal value* is $p^* = \inf\{f_0(x) : x \in \mathcal{F}\}$. A point x^* with $f_0(x^*) = p^*$ is an *optimal point* (or *minimizer*).

Theorem 2.1 (Local optimality implies global optimality)**.** *For the convex program* (1)*, every local minimizer is a global minimizer.*

Proof. Let x^* be a local minimizer: there exists $\delta > 0$ such that $f_0(x^*) \le f_0(x)$ for all feasible x with $\|x - x^*\| \le \delta$. Suppose for contradiction there exists a feasible y with $f_0(y) < f_0(x^*)$. Consider $z_t = (1-t)x^* + ty$ for $t \in (0,1)$. By convexity of each f_i and affinity of the equality constraints, z_t is feasible. Moreover,

$$f_0(z_t) \le (1-t)f_0(x^*) + tf_0(y) < f_0(x^*)$$

for all $t \in (0,1)$. Choosing t small enough that $\|z_t - x^*\| = t\|y - x^*\| < \delta$ contradicts local optimality. □

2.2 Important Special Cases

Definition 2.3 (Linear program – LP)**.** A *linear program* has the form

$$\min_x \ c^\top x \quad \text{s.t.} \quad Ax \le b,\ Cx = d,$$

where all functions are affine (hence convex).

Definition 2.4 (Quadratic program – QP)**.** A *quadratic program* minimizes a convex quadratic over a polyhedral feasible set:

$$\min_x \ \tfrac{1}{2}x^\top Px + q^\top x \quad \text{s.t.} \quad Ax \le b,$$

with $P \succeq 0$ (positive semidefinite).

Definition 2.5 (Second-order cone program – SOCP)**.** An SOCP has constraints of the form $\|A_i x + b_i\|_2 \le c_i^\top x + d_i$ (second-order cone constraints), which are convex.

Definition 2.6 (Semidefinite program – SDP)**.** An SDP minimizes a linear objective subject to a linear matrix inequality:

$$\min_x \ c^\top x \quad \text{s.t.} \quad F_0 + \sum_{i=1}^n x_i F_i \succeq 0,$$

where $F_0, \ldots, F_n$ are symmetric matrices.

Remark 2.1 (Hierarchy). We have the strict inclusion chain: LP $\subset$ QP $\subset$ SOCP $\subset$ SDP $\subset$ convex programs.

RL Connection

In RL, linear programs arise in the LP formulation of MDPs (minimize $\sum_s c(s)V(s)$ subject to Bellman feasibility constraints). Semidefinite programs appear in robust MDP formulations and certain Lyapunov-based safety constraints. Quadratic programs arise in trust-region policy optimization when the surrogate objective is approximated to second order.

3 Lagrangian Duality

3.1 The Lagrangian and Dual Function

Definition 3.1 (Lagrangian)**.** Given the convex program (1), the *Lagrangian* $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ is

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j (a_j^\top x - b_j), \tag{2}$$

where $\lambda \in \mathbb{R}_+^m$ are the *dual variables* (or *Lagrange multipliers*) for the inequality constraints and $\nu \in \mathbb{R}^p$ are the dual variables for the equality constraints.

Definition 3.2 (Lagrange dual function)**.** The *Lagrange dual function* $g : \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R} \cup \{-\infty\}$ is

$$g(\lambda, \nu) = \inf_{x \in \mathbb{R}^n} L(x, \lambda, \nu). \tag{3}$$

Proposition 3.1 (Dual function is concave)**.** *For any optimization problem (not necessarily convex), the dual function g is concave, even when the primal problem is not convex.*

Proof. For each fixed x, the map $(\lambda, \nu) \mapsto L(x, \lambda, \nu)$ is affine (hence concave) in (λ, ν). The pointwise infimum over x of a family of concave functions is concave. □

3.2 Weak Duality

Definition 3.3 (Dual problem)**.** The *Lagrange dual problem* is

$$\max_{\lambda \geq 0,\, \nu} \; g(\lambda, \nu). \tag{4}$$

Its optimal value is denoted d^*.

Theorem 3.2 (Weak duality)**.** *For any optimization problem (not necessarily convex),*

$$d^* \leq p^*. \tag{5}$$

The quantity $p^* - d^*$ is called the duality gap.

Proof. Let $\tilde{x}$ be any feasible point for the primal problem, so $f_i(\tilde{x}) \leq 0$ and $a_j^\top \tilde{x} = b_j$. For any $\lambda \geq 0$ and any ν,

$$L(\tilde{x}, \lambda, \nu) = f_0(\tilde{x}) + \sum_{i=1}^{m} \underbrace{\lambda_i}_{\geq 0} \underbrace{f_i(\tilde{x})}_{\leq 0} + \sum_{j=1}^{p} \nu_j \underbrace{(a_j^\top \tilde{x} - b_j)}_{=0} \leq f_0(\tilde{x}).$$

Taking the infimum over x gives $g(\lambda, \nu) \leq L(\tilde{x}, \lambda, \nu) \leq f_0(\tilde{x})$. Since this holds for every feasible $\tilde{x}$, we get $g(\lambda, \nu) \leq p^*$. Maximizing the left side over (λ, ν) with $\lambda \geq 0$ yields $d^* \leq p^*$. □

3.3 Strong Duality and Slater's Condition

Definition 3.4 (Strong duality)**.** *Strong duality* holds when $d^* = p^*$, i.e., the duality gap is zero.

Definition 3.5 (Slater's condition)**.** A convex program (1) satisfies *Slater's condition* (or *Slater's constraint qualification*) if there exists a point $\hat{x} \in \operatorname{relint}(\operatorname{dom} f_0)$ such that

$$f_i(\hat{x}) < 0, \quad i = 1, \ldots, m, \qquad a_j^\top \hat{x} = b_j, \quad j = 1, \ldots, p.$$

Such $\hat{x}$ is called a *strictly feasible point.*

Theorem 3.3 (Slater implies strong duality)**.** *If the convex program (1) is feasible and Slater's condition holds, then strong duality holds: $d^* = p^*$. Moreover, if p^* is finite, the dual optimum is attained.*

Proof sketch. We use a supporting-hyperplane argument on the set

$$\mathcal{G} = \{(u, v, t) \in \mathbb{R}^m \times \mathbb{R}^p \times \mathbb{R} : \exists\, x \text{ with } f_i(x) \leq u_i,\ a_j^\top x - b_j = v_j,\ f_0(x) \leq t\}.$$

This set is convex (the f_i are convex and $a_j^\top x - b_j$ is affine). The point $(0, 0, p^*)$ lies on the boundary of $\mathcal{G}$ (it is achieved in the limit by a minimizing sequence).

We claim $(0, 0, p^* - \epsilon) \notin \mathcal{G}$ for any $\epsilon > 0$, by definition of p^*. Hence there is a supporting hyperplane at $(0, 0, p^*)$: there exist $(\lambda, \nu, \mu) \neq 0$ with $\mu \geq 0$ such that

$$\lambda^\top u + \nu^\top v + \mu t \geq \mu p^*$$

for all $(u, v, t) \in \mathcal{G}$, with $\lambda \geq 0$.

Claim: $\mu > 0$. Suppose $\mu = 0$. Then $\lambda^\top u + \nu^\top v \geq 0$ for all $(u, v, t) \in \mathcal{G}$. Evaluating at $x = \hat{x}$ (the Slater point) with $u_i = f_i(\hat{x}) < 0$, $v_j = 0$, gives $\sum_i \lambda_i f_i(\hat{x}) \geq 0$. Since $\lambda \geq 0$ and $f_i(\hat{x}) < 0$, we get $\lambda = 0$. Then $\nu^\top v \geq 0$ for all achievable v, which forces $\nu = 0$ (since v can be any value via adjusting x). This contradicts $(\lambda, \nu, \mu) \neq 0$.

With $\mu > 0$, divide through by μ and set $\bar{\lambda} = \lambda/\mu$, $\bar{\nu} = \nu/\mu$. For any feasible x (where $u_i = f_i(x)$, $v_j = 0$, $t = f_0(x)$):

$$\sum_i \bar{\lambda}_i f_i(x) + f_0(x) \geq p^*.$$

Taking the infimum over x gives $g(\bar{\lambda}, \bar{\nu}) \geq p^*$. Combined with weak duality $g \leq p^*$, we conclude $d^* \geq g(\bar{\lambda}, \bar{\nu}) = p^*$, hence $d^* = p^*$. The dual optimum is attained at $(\bar{\lambda}, \bar{\nu})$. □

RL Connection

Strong duality is the theoretical foundation for the Lagrangian approach to constrained MDPs (CMDPs). In a CMDP, we augment the reward objective with penalty terms for constraint violations. Slater's condition guarantees there is no duality gap, so solving the dual (which is unconstrained) is equivalent to solving the primal. This justifies primal-dual policy optimization algorithms such as Lagrangian PPO for safe RL.

4 Karush–Kuhn–Tucker Conditions

Definition 4.1 (KKT conditions)**.** For the convex program (1), the *Karush–Kuhn–Tucker (KKT) conditions* for the primal-dual triple (x^*, λ^*, ν^*) are:

(KKT1) **Primal feasibility:** $f_i(x^*) \leq 0$ for $i = 1, \ldots, m$; $a_j^\top x^* = b_j$ for $j = 1, \ldots, p$.

(KKT2) **Dual feasibility:** $\lambda_i^* \geq 0$ for $i = 1, \ldots, m$.

(KKT3) **Complementary slackness:** $\lambda_i^* f_i(x^*) = 0$ for $i = 1, \ldots, m$.

(KKT4) **Stationarity:** $\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* a_j = 0$.

Theorem 4.1 (KKT necessary conditions under strong duality)**.** *If strong duality holds and x^* is primal optimal, (λ^*, ν^*) is dual optimal, and the f_i are differentiable at x^*, then the KKT conditions hold at (x^*, λ^*, ν^*).*

Proof. Since strong duality holds, $f_0(x^*) = p^* = d^* = g(\lambda^*, \nu^*)$. By definition,

$$g(\lambda^*, \nu^*) = \inf_x L(x, \lambda^*, \nu^*) \leq L(x^*, \lambda^*, \nu^*).$$

We expand:

$$L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \sum_i \lambda_i^* f_i(x^*) + \sum_j \nu_j^* (a_j^\top x^* - b_j).$$

By primal feasibility, $a_j^\top x^* - b_j = 0$ and $f_i(x^*) \leq 0$. Since $\lambda^* \geq 0$, we have $\sum_i \lambda_i^* f_i(x^*) \leq 0$, so $L(x^*, \lambda^*, \nu^*) \leq f_0(x^*)$.

Combining: $f_0(x^*) = g(\lambda^*, \nu^*) \leq L(x^*, \lambda^*, \nu^*) \leq f_0(x^*)$. Hence all inequalities are equalities. In particular, $\sum_i \lambda_i^* f_i(x^*) = 0$. Since each term is non-positive ($\lambda_i^* \geq 0$, $f_i(x^*) \leq 0$), each term is zero: $\lambda_i^* f_i(x^*) = 0$. This gives **complementary slackness**.

Furthermore, $g(\lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*)$, meaning x^* minimizes $L(\cdot, \lambda^*, \nu^*)$ over $\mathbb{R}^n$. Since this is an unconstrained minimum of a differentiable function, the gradient vanishes:

$$\nabla_x L(x^*, \lambda^*, \nu^*) = \nabla f_0(x^*) + \sum_i \lambda_i^* \nabla f_i(x^*) + \sum_j \nu_j^* a_j = 0.$$

This is **stationarity**. Primal and dual feasibility hold by assumption. □

Theorem 4.2 (KKT sufficient conditions for convex problems)**.** *For the convex program (1) with differentiable $f_0, \ldots, f_m$, if a triple (x^*, λ^*, ν^*) satisfies the KKT conditions, then x^* is primal optimal and (λ^*, ν^*) is dual optimal, with zero duality gap.*

Proof. By stationarity, x^* minimizes $L(\cdot, \lambda^*, \nu^*)$ (since L is convex in x and the gradient vanishes at x^*). Therefore,

$$g(\lambda^*, \nu^*) = \inf_x L(x, \lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*).$$

By complementary slackness and primal feasibility,

$$L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \underbrace{\sum_i \lambda_i^* f_i(x^*)}_{=0} + \underbrace{\sum_j \nu_j^* (a_j^\top x^* - b_j)}_{=0} = f_0(x^*).$$

Hence $g(\lambda^*, \nu^*) = f_0(x^*)$. By weak duality, $d^* \geq g(\lambda^*, \nu^*) = f_0(x^*) \geq p^* \geq d^*$, so all are equal. Thus x^* is primal optimal and (λ^*, ν^*) is dual optimal. □

KKT Conditions for Convex Problems

For convex programs with differentiable objectives and constraints, the KKT conditions are **necessary** (assuming a constraint qualification such as Slater's condition, which implies strong duality) and **sufficient** for optimality. This makes them the central tool for characterizing solutions.

RL Connection

The KKT conditions characterize solutions of the policy optimization subproblem in TRPO and PPO. When we constrain the policy update to a KL trust region, the KKT multiplier on the KL constraint determines the effective step size. Complementary slackness tells us: either the constraint is active (the update reaches the trust-region boundary) or the multiplier is zero (the unconstrained optimum lies inside the trust region).

5 Gradient Descent

5.1 Smoothness and Convexity

Definition 5.1 (L-smoothness)**.** A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is *L-smooth* (or has L-Lipschitz continuous gradient) if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \forall x, y \in \mathbb{R}^n. \tag{6}$$

Lemma 5.1 (Descent lemma)**.** *If f is L-smooth, then for all $x, y \in \mathbb{R}^n$:*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2. \tag{7}$$

Proof. By the fundamental theorem of calculus,

$$f(y) - f(x) = \int_0^1 \nabla f(x + t(y - x))^\top (y - x)\, dt.$$

Subtracting $\nabla f(x)^\top (y - x)$:

$$\begin{aligned} f(y) - f(x) - \nabla f(x)^\top (y - x) &= \int_0^1 [\nabla f(x + t(y - x)) - \nabla f(x)]^\top (y - x)\, dt \\ &\leq \int_0^1 \|\nabla f(x + t(y - x)) - \nabla f(x)\| \cdot \|y - x\|\, dt \\ &\leq \int_0^1 Lt\|y - x\|^2\, dt = \frac{L}{2}\|y - x\|^2. \end{aligned}$$

□

Definition 5.2 (μ-strong convexity)**.** A differentiable function f is *μ-strongly convex* (with $\mu > 0$) if for all x, y:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2. \tag{8}$$

Remark 5.1. If f is μ-strongly convex and L-smooth, the *condition number* is $\kappa = L/\mu \geq 1$. The condition number governs the convergence speed of first-order methods.

5.2 Gradient Descent Algorithm

The gradient descent iteration with step size $\eta > 0$ is:

$$x_{k+1} = x_k - \eta \nabla f(x_k). \tag{9}$$

5.3 Convergence for Convex Functions

Theorem 5.2 ($O(1/k)$ convergence for convex L-smooth functions)**.** *Let f be convex and L-smooth with minimizer x^*. Gradient descent with step size $\eta = 1/L$ satisfies*

$$f(x_k) - f(x^*) \leq \frac{L\|x_0 - x^*\|^2}{2k}. \tag{10}$$

Proof. Set $\eta = 1/L$. By the descent lemma (Lemma 5.1) with $y = x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k)$:

$$f(x_{k+1}) \leq f(x_k) + \nabla f(x_k)^\top (x_{k+1} - x_k) + \frac{L}{2}\|x_{k+1} - x_k\|^2 = f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2. \tag{11}$$

So the sequence of function values is non-increasing, and

$$f(x_k) - f(x_{k+1}) \geq \frac{1}{2L}\|\nabla f(x_k)\|^2. \tag{12}$$

By convexity, $f(x^*) \geq f(x_k) + \nabla f(x_k)^\top (x^* - x_k)$, hence

$$f(x_k) - f(x^*) \leq \nabla f(x_k)^\top (x_k - x^*) \leq \|\nabla f(x_k)\| \cdot \|x_k - x^*\|. \tag{13}$$

Now consider the distance to the optimum:

$$\begin{aligned} \|x_{k+1} - x^*\|^2 &= \|x_k - \tfrac{1}{L}\nabla f(x_k) - x^*\|^2 \\ &= \|x_k - x^*\|^2 - \frac{2}{L}\nabla f(x_k)^\top (x_k - x^*) + \frac{1}{L^2}\|\nabla f(x_k)\|^2. \end{aligned} \tag{14}$$

Using convexity: $\nabla f(x_k)^\top (x_k - x^*) \geq f(x_k) - f(x^*)$, so

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - \frac{2}{L}(f(x_k) - f(x^*)) + \frac{1}{L^2}\|\nabla f(x_k)\|^2.$$

From (12): $\|\nabla f(x_k)\|^2 \leq 2L(f(x_k) - f(x_{k+1}))$, so

$$\begin{aligned} \|x_{k+1} - x^*\|^2 &\leq \|x_k - x^*\|^2 - \frac{2}{L}(f(x_k) - f(x^*)) + \frac{2}{L}(f(x_k) - f(x_{k+1})) \\ &= \|x_k - x^*\|^2 - \frac{2}{L}(f(x_{k+1}) - f(x^*)). \end{aligned}$$

Rearranging: $f(x_{k+1}) - f(x^*) \leq \frac{L}{2}(\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2)$.

Let $\delta_k = f(x_k) - f(x^*)$. Summing from $k = 0$ to $k = K - 1$:

$$\sum_{k=1}^{K} \delta_k \leq \frac{L}{2}\|x_0 - x^*\|^2.$$

Since δ_k is non-increasing (from (12)), $K\delta_K \leq \sum_{k=1}^{K} \delta_k \leq \frac{L}{2}\|x_0 - x^*\|^2$, giving $\delta_K \leq \frac{L\|x_0 - x^*\|^2}{2K}$. □

5.4 Convergence for Strongly Convex Functions

Theorem 5.3 (Linear convergence for strongly convex L-smooth functions)**.** *Let f be μ-strongly convex and L-smooth. Gradient descent with $\eta = 1/L$ satisfies*

$$f(x_k) - f(x^*) \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f(x^*)). \tag{15}$$

Hence $f(x_k) - f(x^*) \leq \epsilon$ after $k = O(\kappa \log(1/\epsilon))$ iterations, where $\kappa = L/\mu$.

Proof. From (11), $f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2$. By strong convexity applied at the optimum x^*:

$$f(x^*) \geq f(x_k) + \nabla f(x_k)^\top (x^* - x_k) + \frac{\mu}{2}\|x^* - x_k\|^2.$$

Thus $f(x_k) - f(x^*) \leq \nabla f(x_k)^\top (x_k - x^*) - \frac{\mu}{2}\|x_k - x^*\|^2$. By the Cauchy–Schwarz inequality and the AM-GM inequality (or by optimizing over $\|x_k - x^*\|$), one obtains the standard inequality for strongly convex functions:

$$\|\nabla f(x_k)\|^2 \geq 2\mu(f(x_k) - f(x^*)). \tag{16}$$

This is the *Polyak–Lojasiewicz (PL) inequality*, which follows from strong convexity.

Proof of (16)*:* By strong convexity, for all y: $f(y) \geq f(x_k) + \nabla f(x_k)^\top (y - x_k) + \frac{\mu}{2}\|y - x_k\|^2$. Minimizing the right side over y (a quadratic in y) gives $y^* = x_k - \frac{1}{\mu}\nabla f(x_k)$, yielding

$$f(x^*) \geq f(x_k) - \frac{1}{2\mu}\|\nabla f(x_k)\|^2,$$

which rearranges to (16).

Substituting (16) into the descent bound:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L} \cdot 2\mu(f(x_k) - f(x^*)) = f(x_k) - \frac{\mu}{L}(f(x_k) - f(x^*)).$$

Subtracting $f(x^*)$ from both sides:

$$f(x_{k+1}) - f(x^*) \leq \left(1 - \frac{\mu}{L}\right)(f(x_k) - f(x^*)).$$

Applying this recursively yields (15) with $\rho = 1 - \mu/L = 1 - 1/\kappa$. □

Gradient Descent Convergence Rates

For L-smooth functions with step size $\eta = 1/L$:

- **Convex:** $f(x_k) - f^* = O(1/k)$, i.e., $O(1/\epsilon)$ iterations for ϵ-accuracy.
- **μ-strongly convex:** $f(x_k) - f^* = O(\rho^k)$ with $\rho = 1 - 1/\kappa$, i.e., $O(\kappa \log(1/\epsilon))$ iterations.

RL Connection

Policy gradient methods and TD learning with function approximation perform (stochastic) gradient descent on the policy parameters or value function weights. The convergence rate depends critically on the smoothness (L) and strong convexity (μ) of the loss landscape. In practice, the loss for neural network policies is non-convex, but locally the landscape often behaves like a strongly convex function near a good solution.

6 Nesterov's Accelerated Gradient Method

6.1 The Accelerated Iteration

Nesterov's accelerated gradient method improves the $O(1/k)$ rate for convex functions to $O(1/k^2)$, which is optimal among first-order methods.

Definition 6.1 (Nesterov's accelerated gradient method)**.** For a convex, L-smooth function f, with step size $\eta = 1/L$, the accelerated gradient method maintains two sequences:

$$\begin{aligned} y_k &= x_k + \frac{k-1}{k+2}(x_k - x_{k-1}), \\ x_{k+1} &= y_k - \frac{1}{L}\nabla f(y_k), \end{aligned} \tag{17}$$

with $x_0 = x_{-1}$ (so $y_0 = x_0$). The momentum coefficient $\theta_k = (k-1)/(k+2)$ is the key ingredient.

6.2 Convergence Rate

Theorem 6.1 ($O(1/k^2)$ convergence of Nesterov's method)**.** *Under the conditions of Definition 6.1,*

$$f(x_k) - f(x^*) \leq \frac{2L\|x_0 - x^*\|^2}{(k+1)^2}. \tag{18}$$

Proof. We use the *estimate sequence* technique. Define the sequences $\{A_k\}$ and $\{v_k\}$ by $A_0 = 0$, $a_{k+1} = (k+1)/2$, $A_{k+1} = A_k + a_{k+1}$, and $v_0 = x_0$. One can verify that $A_k = k(k+1)/4$ for $k \geq 1$.

We prove by induction the key inequality:

$$A_k(f(x_k) - f(x^*)) \leq \frac{L}{2}\|v_0 - x^*\|^2 - \frac{L}{2}\|v_k - x^*\|^2. \tag{19}$$

Since $\|v_k - x^*\|^2 \geq 0$, this implies $f(x_k) - f(x^*) \leq \frac{L\|x_0 - x^*\|^2}{2A_k} = \frac{2L\|x_0 - x^*\|^2}{k(k+1)} \leq \frac{2L\|x_0 - x^*\|^2}{(k+1)^2}$.

Base case ($k = 0$): $A_0 = 0$, so both sides equal zero.

Inductive step: Assume (19) holds for k. We need to show it for $k+1$. By the descent lemma applied at y_k:

$$f(x_{k+1}) \leq f(y_k) - \frac{1}{2L}\|\nabla f(y_k)\|^2.$$

By convexity: $f(x^*) \geq f(y_k) + \nabla f(y_k)^\top(x^* - y_k)$, so $f(y_k) \leq f(x^*) + \nabla f(y_k)^\top(y_k - x^*)$. Combining with a weighted average (weight a_{k+1} for the inequality at x^* and A_k for the inductive hypothesis at x_k), and using the specific choice $y_k = \frac{A_k x_k + a_{k+1} v_k}{A_{k+1}}$ together with the update $v_{k+1} = v_k - \frac{a_{k+1}}{L}\nabla f(y_k)$, one obtains after algebraic manipulations (completing the square in v_k):

$$A_{k+1}(f(x_{k+1}) - f(x^*)) \leq \frac{L}{2}\|v_0 - x^*\|^2 - \frac{L}{2}\|v_{k+1} - x^*\|^2.$$

The algebraic details involve verifying that the momentum coefficient $(k-1)/(k+2)$ corresponds precisely to the ratio A_k/A_{k+1} that makes y_k the correct convex combination. This completes the induction. □

Remark 6.1 (Optimality). Nesterov (1983) showed that $O(1/k^2)$ is the *optimal* rate for first-order methods on the class of L-smooth convex functions. That is, for any first-order method using only gradient evaluations, there exists an L-smooth convex function for which the method cannot converge faster than $\Omega(1/k^2)$. This is the celebrated information-theoretic lower bound.

Warning

Nesterov acceleration does not directly apply to stochastic gradients. Naive momentum with noisy gradients can amplify variance and diverge. This is why SGD with momentum and Adam use different momentum schemes than classical Nesterov acceleration.

7 Proximal Methods

7.1 The Proximal Operator

Many problems in ML involve non-smooth regularizers (e.g., ℓ_1 for sparsity). Proximal methods extend gradient descent to handle such terms.

Definition 7.1 (Proximal operator)**.** For a closed convex function $h : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$, the *proximal operator* $\mathrm{prox}_h : \mathbb{R}^n \to \mathbb{R}^n$ is

$$\mathrm{prox}_h(v) = \arg\min_{x \in \mathbb{R}^n} \left\{ h(x) + \frac{1}{2}\|x - v\|^2 \right\}. \tag{20}$$

Proposition 7.1 (Well-definedness)**.** *For any closed convex h with $\mathrm{dom}\, h \neq \emptyset$, the proximal operator $\mathrm{prox}_h(v)$ exists and is unique for every $v \in \mathbb{R}^n$.*

Proof. The objective $h(x) + \frac{1}{2}\|x - v\|^2$ is strongly convex (with $\mu = 1$), hence has at most one minimizer. It is also coercive (goes to $+\infty$ as $\|x\| \to \infty$) and lower semicontinuous, so a minimizer exists. □

Example 7.1 (ℓ_1 proximal: soft thresholding)**.** For $h(x) = \tau\|x\|_1$ with $\tau > 0$, the proximal operator is the soft-thresholding operator applied componentwise:

$$[\mathrm{prox}_{\tau\|\cdot\|_1}(v)]_i = \mathrm{sign}(v_i)\max(|v_i| - \tau, 0).$$

Example 7.2 (Indicator function)**.** For $h = \iota_C$ (the indicator function of a closed convex set C, equal to 0 on C and $+\infty$ elsewhere), $\mathrm{prox}_h(v) = \Pi_C(v)$, the Euclidean projection onto C.

7.2 Proximal Gradient Method

Consider the composite minimization problem:

$$\min_{x \in \mathbb{R}^n} F(x) = f(x) + h(x), \tag{21}$$

where f is convex and L-smooth, and h is convex but possibly non-smooth.

Definition 7.2 (Proximal gradient method)**.** The *proximal gradient method* with step size $\eta = 1/L$ is:

$$x_{k+1} = \mathrm{prox}_{\eta h}\big(x_k - \eta \nabla f(x_k)\big). \tag{22}$$

Theorem 7.2 (Convergence of proximal gradient)**.** *Under the composite model (21) with f convex and L-smooth, the proximal gradient method with $\eta = 1/L$ achieves*

$$F(x_k) - F(x^*) \leq \frac{L\|x_0 - x^*\|^2}{2k}.$$

The convergence rate is $O(1/k)$, matching gradient descent.

Proof. The key observation is that the proximal gradient step can be written as $x_{k+1} = \arg\min_x\{f(x_k) + \nabla f(x_k)^\top(x - x_k) + \frac{L}{2}\|x - x_k\|^2 + h(x)\}$. By the descent lemma, $f(x_{k+1}) \le f(x_k) + \nabla f(x_k)^\top(x_{k+1} - x_k) + \frac{L}{2}\|x_{k+1} - x_k\|^2$. Adding $h(x_{k+1})$ to both sides:

$$F(x_{k+1}) \le f(x_k) + \nabla f(x_k)^\top(x_{k+1} - x_k) + \frac{L}{2}\|x_{k+1} - x_k\|^2 + h(x_{k+1}).$$

By the optimality condition for the proximal step, for any x:

$$f(x_k) + \nabla f(x_k)^\top(x_{k+1} - x_k) + \frac{L}{2}\|x_{k+1} - x_k\|^2 + h(x_{k+1}) \le f(x_k) + \nabla f(x_k)^\top(x - x_k) + \frac{L}{2}\|x - x_k\|^2 + h(x).$$

Setting $x = x^*$ and using convexity of f ($f(x_k) + \nabla f(x_k)^\top(x^* - x_k) \le f(x^*)$):

$$F(x_{k+1}) \le F(x^*) + \frac{L}{2}\|x^* - x_k\|^2.$$

A telescoping argument identical to the one in Theorem 5.2 (tracking $\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2$) yields the result. □

7.3 FISTA: Fast Iterative Shrinkage-Thresholding Algorithm

Definition 7.3 (FISTA)**.** FISTA applies Nesterov acceleration to the proximal gradient method:

$$\begin{aligned} y_k &= x_k + \frac{k-1}{k+2}(x_k - x_{k-1}), \\ x_{k+1} &= \operatorname{prox}_{\eta h}\big(y_k - \eta \nabla f(y_k)\big), \end{aligned} \tag{23}$$

with $\eta = 1/L$ and $x_0 = x_{-1}$.

Theorem 7.3 (FISTA convergence)**.** *Under the composite model (21), FISTA achieves*

$$F(x_k) - F(x^*) \le \frac{2L\|x_0 - x^*\|^2}{(k+1)^2}.$$

This is the optimal $O(1/k^2)$ rate for first-order methods on this problem class.

Proof. The proof follows the estimate-sequence technique of Theorem 6.1, replacing gradient steps with proximal gradient steps. The descent lemma for the smooth part f and the convexity of h allow the same inductive argument to go through with $F = f + h$ in place of f. □

RL Connection

Proximal methods are used in RL for imposing structure on learned policies. For example, ℓ_1-regularized policy optimization promotes sparse feature representations. The mirror descent interpretation of proximal methods (with Bregman divergences replacing the Euclidean distance) connects to natural policy gradient, where the KL divergence plays the role of the Bregman divergence in policy space.

8 Interior-Point Methods

8.1 Logarithmic Barrier

Interior-point methods solve constrained problems by converting inequality constraints into a smooth barrier term added to the objective, then following the "central path" as the barrier weight decreases.

Definition 8.1 (Logarithmic barrier)**.** For the convex program (1) (with equality constraints omitted for simplicity), the *logarithmic barrier* is

$$\phi(x) = -\sum_{i=1}^{m} \log(-f_i(x)), \tag{24}$$

defined for x in the strictly feasible set $\{x : f_i(x) < 0,\ i = 1, \ldots, m\}$.

Proposition 8.1 (Barrier is convex)**.** *If $f_1, \ldots, f_m$ are convex, then the logarithmic barrier ϕ is convex on $\{x : f_i(x) < 0\}$.*

Proof. The function $-\log(-u)$ is convex and non-decreasing on $u < 0$. Since each f_i is convex, the composition $-\log(-f_i(x))$ is convex (by the composition rule for convex non-decreasing outer function and convex inner function). The sum of convex functions is convex. □

8.2 Central Path

Definition 8.2 (Barrier problem and central path)**.** For parameter $t > 0$, the *barrier problem* is

$$\min_x \ t\, f_0(x) + \phi(x). \tag{25}$$

Assuming a unique minimizer $x^*(t)$ exists for each $t > 0$, the curve $\{x^*(t) : t > 0\}$ is called the *central path.*

Theorem 8.2 (Central path optimality bound)**.** *Each point $x^*(t)$ on the central path satisfies*

$$f_0(x^*(t)) - p^* \leq \frac{m}{t}, \tag{26}$$

where p^* is the optimal value and m is the number of inequality constraints.

Proof. At $x^*(t)$, the gradient of the barrier objective vanishes:

$$t\,\nabla f_0(x^*(t)) + \nabla\phi(x^*(t)) = 0.$$

The gradient of the barrier is $\nabla\phi(x) = \sum_{i=1}^{m} \frac{-\nabla f_i(x)}{-f_i(x)} = \sum_{i=1}^{m} \frac{\nabla f_i(x)}{-f_i(x)}$.

Define the dual variables $\lambda_i^*(t) = \frac{1}{t(-f_i(x^*(t)))} > 0$. Then the stationarity condition becomes

$$\nabla f_0(x^*(t)) + \sum_{i=1}^{m} \lambda_i^*(t) \nabla f_i(x^*(t)) = 0.$$

Thus $x^*(t)$ minimizes $L(x, \lambda^*(t), 0) = f_0(x) + \sum_i \lambda_i^*(t) f_i(x)$ over x, so the dual function evaluates to

$$\begin{aligned} g(\lambda^*(t), 0) &= f_0(x^*(t)) + \sum_i \lambda_i^*(t) f_i(x^*(t)) \\ &= f_0(x^*(t)) + \sum_i \frac{f_i(x^*(t))}{t(-f_i(x^*(t)))} \\ &= f_0(x^*(t)) - \frac{m}{t}. \end{aligned}$$

By weak duality, $g(\lambda^*(t), 0) \leq p^*$, so $f_0(x^*(t)) - m/t \leq p^*$, giving $f_0(x^*(t)) - p^* \leq m/t$. □

8.3 Barrier Method

Definition 8.3 (Barrier method)**.** The *barrier method* (or *path-following method*) proceeds as follows:

1. Initialize with a strictly feasible $x^{(0)}$, $t^{(0)} > 0$, and a growth factor $\sigma > 1$.
2. At iteration ℓ: solve (approximately) the barrier problem $\min_x \; t^{(\ell)} f_0(x) + \phi(x)$ starting from $x^{(\ell)}$, using Newton's method. Set $x^{(\ell+1)}$ to the result.
3. Update $t^{(\ell+1)} = \sigma \cdot t^{(\ell)}$.
4. Stop when $m/t^{(\ell)} \leq \epsilon$.

Theorem 8.3 (Complexity of the barrier method)**.** *To achieve $f_0(x) - p^* \leq \epsilon$, the barrier method requires $O(\sqrt{m}\log(m/\epsilon))$ Newton steps (outer iterations), each involving the solution of a linear system of size $n \times n$.*

Proof sketch. By Theorem 8.2, we need $t \geq m/\epsilon$, so $\lceil \log_\sigma(m/(t^{(0)}\epsilon)) \rceil$ outer iterations suffice. The key result from the theory of self-concordant barriers (Nesterov and Nemirovskii) is that with the optimal choice $\sigma = 1 + 1/\sqrt{m}$, each centering step (finding $x^*(t)$ to sufficient accuracy) requires $O(1)$ Newton steps (specifically, a constant number independent of m and n). The total number of outer iterations is then $\log_\sigma(m/\epsilon) = O(\sqrt{m}\log(m/\epsilon))$, using $\log(1 + 1/\sqrt{m}) \approx 1/\sqrt{m}$. □

Interior-Point Complexity

Interior-point methods solve convex programs with m inequality constraints to ϵ-accuracy in $O(\sqrt{m}\log(m/\epsilon))$ Newton iterations. Each Newton iteration costs $O(n^3)$ for a dense n-dimensional problem (or less with structure). This polynomial complexity in m, n, and $\log(1/\epsilon)$ makes interior-point methods highly efficient for moderate-dimensional problems.

RL Connection

Interior-point methods are used to solve the linear programs that arise in the LP formulation of finite MDPs. For large-scale problems, they complement value iteration and policy iteration. In constrained RL (safe RL), solving the inner constrained optimization at each policy update step can be handled by interior-point methods when the constraint set has moderate dimension.

9 Exercises

Exercise 9.1 (Dual of a linear program)**.** Consider the LP $\min_x \; c^\top x$ subject to $Ax = b$, $x \geq 0$.

(a) Write the Lagrangian and derive the dual problem. Show it is $\max_\nu \; b^\top \nu$ subject to $A^\top \nu \leq c$.

(b) Prove that strong duality always holds for feasible LPs (Hint: LP feasible sets are polyhedral, so Slater's condition can be replaced by the weaker condition of feasibility).

Exercise 9.2 (KKT for least-squares with non-negativity)**.** Consider the non-negative least-squares problem: $\min_{x \geq 0} \; \frac{1}{2}\|Ax - b\|^2$. Write the KKT conditions explicitly and show they reduce to: $x \geq 0$, $A^\top(Ax - b) \geq 0$, $x_i[A^\top(Ax - b)]_i = 0$ for all i.

Exercise 9.3 (Convergence rate verification)**.** Let $f(x) = \frac{1}{2}x^\top Q x$ where Q is symmetric positive definite with eigenvalues $\mu = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$.

(a) Verify that f is μ-strongly convex and L-smooth.

(b) Run gradient descent with $\eta = 1/L$ and verify that $f(x_k) \leq (1 - \mu/L)^k f(x_0)$.

(c) For $Q = \mathrm{diag}(1, 100)$, compute the condition number and determine how many iterations are needed for $f(x_k)/f(x_0) \leq 10^{-6}$.

Exercise 9.4 (Soft-thresholding derivation)**.** Derive the soft-thresholding operator: show that $\mathrm{prox}_{\tau|\cdot|}(v) = \mathrm{sign}(v)\max(|v| - \tau, 0)$ for a scalar v. (Hint: consider the three cases $v > \tau$, $|v| \leq \tau$, and $v < -\tau$ separately.)

Exercise 9.5 (Dual of SDP relaxation)**.** Consider the boolean optimization problem $\min_{x \in \{-1,+1\}^n} x^\top W x$. The standard SDP relaxation replaces $xx^\top$ with a matrix variable $X \succeq 0$, $X_{ii} = 1$.

(a) Write this relaxation as a standard SDP.

(b) Derive its Lagrange dual.

(c) Show that weak duality provides a lower bound on the original combinatorial problem.

Exercise 9.6 (Proving the PL inequality)**.** Let f be μ-strongly convex and differentiable.

(a) Prove that $\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f(x^*))$ for all x.

(b) Show by example that the PL inequality can hold for non-convex functions (consider $f(x) = x^2 + 3\sin^2(x)$ near the origin; verify numerically).

(c) Prove that gradient descent converges linearly under the PL inequality alone (without assuming convexity).

Exercise 9.7 (FISTA for LASSO)**.** The LASSO problem is $\min_x \frac{1}{2}\|Ax - b\|^2 + \tau\|x\|_1$.

(a) Identify $f(x)$ and $h(x)$ in the composite formulation.

(b) Write out the FISTA iteration explicitly (including the soft-thresholding step).

(c) What is the Lipschitz constant L of ∇f? Express it in terms of A.

Exercise 9.8 (Central path for LP)**.** For the LP $\min\ c^\top x$ subject to $x \geq 0$ (i.e., $f_i(x) = -x_i$):

(a) Write the barrier function $\phi(x) = -\sum_i \log(x_i)$.

(b) Show that the central path satisfies $x_i^*(t) = 1/(tc_i)$ when $c > 0$.

(c) Verify that $c^\top x^*(t) - p^* = n/t$ where n is the dimension and $p^* = 0$.

Exercise 9.9 (Duality gap in constrained MDP)**.** Consider a constrained MDP with one constraint: $\max_\pi J(\pi)$ subject to $C(\pi) \leq \alpha$, where J and C are the expected return and expected cost under policy π over a finite MDP.

(a) Write the Lagrangian $L(\pi, \lambda)$ for this problem.

(b) Show that the dual function $g(\lambda) = \max_\pi L(\pi, \lambda)$ corresponds to solving an unconstrained MDP with modified reward $r(s, a) - \lambda c(s, a)$.

(c) Under what condition does strong duality hold? (Hint: relate to Slater's condition in the occupancy-measure LP formulation.)

Exercise 9.10 (Moreau envelope and gradient)**.** The *Moreau envelope* of a convex function h with parameter $\gamma > 0$ is $M_\gamma h(v) = \min_x\{h(x) + \frac{1}{2\gamma}\|x - v\|^2\}$.

(a) Show that $M_\gamma h$ is differentiable even when h is not, with gradient $\nabla M_\gamma h(v) = \frac{1}{\gamma}(v - \mathrm{prox}_{\gamma h}(v))$.

(b) Prove that $M_\gamma h$ is $(1/\gamma)$-smooth.

(c) Explain how this connects to the proximal point algorithm: $x_{k+1} = \mathrm{prox}_{\gamma h}(x_k)$.

10 Summary

10.1 Key Theorems Recap

The central results of this document are:

1. **Weak duality** (Theorem 3.2): Always $d^* \leq p^*$.
2. **Strong duality** (Theorem 3.3): Under Slater's condition, $d^* = p^*$.
3. **KKT conditions** (Theorems 4.1 and 4.2): Necessary under strong duality, sufficient for convex problems.
4. **Gradient descent** (Theorems 5.2 and 5.3): $O(1/k)$ for convex, $O(\rho^k)$ for strongly convex.
5. **Nesterov acceleration** (Theorem 6.1): Optimal $O(1/k^2)$ for smooth convex.
6. **Interior-point methods** (Theorem 8.3): $O(\sqrt{m}\log(m/\epsilon))$ Newton steps.

10.2 Convergence Rate Summary

Table 1: Convergence rates for first-order and interior-point methods.

Method	**Assumption**	**Rate**	**Iterations for ϵ**
Gradient descent	Convex, L-smooth	$O(L\|x_0 - x^*\|^2/k)$	$O(1/\epsilon)$
Gradient descent	μ-s.c., L-smooth	$O((1-\mu/L)^k)$	$O(\kappa\log(1/\epsilon))$
Nesterov accel.	Convex, L-smooth	$O(L\|x_0 - x^*\|^2/k^2)$	$O(1/\sqrt{\epsilon})$
Nesterov accel.	μ-s.c., L-smooth	$O((1-1/\sqrt{\kappa})^k)$	$O(\sqrt{\kappa}\log(1/\epsilon))$
Prox. gradient	Convex, L-smooth + h	$O(L\|x_0 - x^*\|^2/k)$	$O(1/\epsilon)$
FISTA	Convex, L-smooth + h	$O(L\|x_0 - x^*\|^2/k^2)$	$O(1/\sqrt{\epsilon})$
Interior point	m ineq. constraints	$O(\sqrt{m}\log(m/\epsilon))$	$O(\sqrt{m}\log(1/\epsilon))$

Remark 10.1 (Lower bounds). The $O(1/k^2)$ rate for smooth convex functions and $O((1-1/\sqrt{\kappa})^k)$ for smooth strongly convex functions match the information-theoretic lower bounds for first-order methods (Nesterov, 1983). These are not improvable by any method that accesses f only through gradient evaluations.

Remark 10.2 (Stochastic extensions). In ML and RL, we typically have access only to stochastic gradients. SGD with step size $\eta_k = O(1/k)$ achieves $O(1/\sqrt{k})$ for convex and $O(1/k)$ for strongly convex problems (in expectation). Variance reduction techniques (SVRG, SARAH) recover the deterministic rates up to logarithmic factors. These stochastic methods are covered in a subsequent document.